

Separate modal analysis for tumor detection with a digital image elasto tomography (DIET) breast cancer screening system

Amer S. Kashif^{a)} and Thomas F. Lotz

Department of Mechanical Engineering, Centre for Bioengineering, University of Canterbury, Christchurch 8041, New Zealand

Adrianus M. W. Heeren

Department of Mechanical Engineering, Eindhoven University of Technology, Eindhoven 5612, The Netherlands

James G. Chase

Department of Mechanical Engineering, Centre for Bioengineering, University of Canterbury, Christchurch 8041, New Zealand

(Received 26 June 2013; revised 3 October 2013; accepted for publication 5 October 2013; published 24 October 2013)

Purpose: It is estimated that every year, 1×10^6 women are diagnosed with breast cancer, and more than 410 000 die annually worldwide. Digital Image Elasto Tomography (DIET) is a new noninvasive breast cancer screening modality that induces mechanical vibrations in the breast and images its surface motion with digital cameras to detect changes in stiffness. This research develops a new automated approach for diagnosing breast cancer using DIET based on a modal analysis model.

Methods: The first and second natural frequency of silicone phantom breasts is analyzed. Separate modal analysis is performed for each region of the phantom to estimate the modal parameters using imaged motion data over several input frequencies. Statistical methods are used to assess the likelihood of a frequency shift, which can indicate tumor location. Phantoms with 5, 10, and 20 mm stiff inclusions are tested, as well as a homogeneous (healthy) phantom. Inclusions are located at four locations with different depth.

Results: The second natural frequency proves to be a reliable metric with the potential to clearly distinguish lesion like inclusions of different stiffness, as well as providing an approximate location for the tumor like inclusions. The 10 and 20 mm inclusions are always detected regardless of depth. The 5 mm inclusions are only detected near the surface. The homogeneous phantom always yields a negative result, as expected.

Conclusions: Detection is based on a statistical likelihood analysis to determine the presence of significantly different frequency response over the phantom, which is a novel approach to this problem. The overall results show promise and justify proof of concept trials with human subjects.

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Key words: breast cancer, digital image elasto tomography, modal analysis, mechanical breast model, elastography, natural frequency

1. INTRODUCTION

Breast cancer is a major international public health issue. Over 1×10^6 women are estimated to be diagnosed every year, with more than 400 000 mortalities.¹ Statistics show that early detection and intervention greatly increases five year survival rates to over 95%,^{2,3} stressing the need for effective wide-scale screening programs. X-ray mammography is currently the only approved large scale breast screening modality, and is widely used in most developed countries. However, mammography is controversial due to low compliance, poor sensitivity,⁴ accumulated radiation exposure,⁵ discomfort due to breast compression,⁶ and overdiagnosis.⁷ Furthermore, breast cancer screening programs are often not available in developing countries due to relatively high costs involved⁸ and rural populations with limited access to specialized radiology clinics.⁹ Hence, requiring a low cost and

less invasive modality is required to better meet screening demands.

Digital Image Elasto Tomography (DIET) is a novel breast cancer screening approach to detect cancerous lesions based on the elastic properties of the breast. This technology is noninvasive, does not require breast compression, is very portable, requires low operator skill (software-based diagnosis and no radiation), and is low in operational cost,^{10–13} rendering this approach potentially very suitable for large screening programs, particularly in remote areas or countries with limited access to screening. The DIET makes use of the high contrast difference in elastic properties of 300%–1500% between cancerous tissue and healthy fatty or fibroglandular tissue in the breast,¹⁴ compared to a 5%–10% radio-density contrast imaged in x-ray mammography.³ DIET applies sinusoidal mechanical vibrations to the breast and the resulting 3D surface oscillations are captured by an array of five

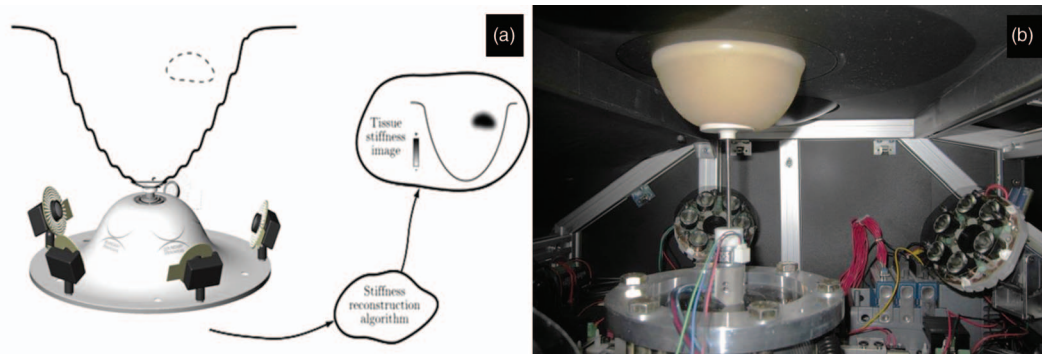


FIG. 1. (a) Conceptual schematic of the imaging principle in the DIET system and (b) inside view of the DIET clinical prototype showing two cameras, actuator, and a breast phantom.

cameras surrounding the breast, each rigidly fixed 72° apart from each other. The cameras are positioned such that each point on the breast surface is captured by at least two cameras to ensure translation of surface motion in 3D. The sinusoidal vibrations are induced by a motion actuator in the nipple area and the surface vibration response is analyzed to detect disturbances in the observed vibration patterns, indicating a higher stiffness area within the healthy soft tissues.^{15–17} These high stiffness areas or inclusions may be considered potentially diagnostic of cancerous lesions. Figure 1(a) shows the conceptual schematic of the imaging principle in the DIET system and Fig. 1(b) shows the inside view of the DIET clinical prototype.

Finite element (FEM) (Ref. 17) or surface motion based methods¹³ can be used to detect the presence, estimated size and location of a tumor like inclusion. FEM can provide a full 3D map of the breast tissue stiffness, but is computationally expensive due to the need for complex, ill-conditioned inverse problem solutions.¹⁸ Surface based methods¹³ show promising results, but rely on a visual inspection of the surface response at different imaging frequencies. Thus, both methods involve an added time cost and require a more expert operator to process and analyze the results.

The minimal modeling method presented in Ref. 19 uses a broader range of imaging frequencies and fits a simple mechanical model to the frequency response and first natural frequency of the breast surface, providing a software based diagnosis that is fully automated. This specific concept seemed to be reliable for shallow inclusions of larger than 20 mm in diameter. However, that analysis was inconclusive for smaller inclusions.

This research presents a concept to detect the smaller inclusions, using the principles of modal analysis, where the frequency response of a system with multiple degrees of freedom (MDOF) is considered to be the summation of the frequency responses corresponding to each of its modes. Hypothesis testing was used to detect significant changes in the second natural frequency across different locations or regions of the breast as being indicative of the presence of inclusion in that region. The objective was set to detect a inclusion size of at least 10 mm, as this value is clinically significant in current screening applications,²⁰ and is the average detection size in many screening studies.^{21,22}

2. MATERIALS AND METHODS

2.A. Experimental setup and data acquisition

This study was performed using a set of 13 anthropomorphic silicone (A-341 Silicone, Factor2, Lakeside) breast phantoms, comprising one homogenous healthy with no inclusion, and 12 heterogeneous phantoms each with a single stiff inclusion of various sizes (5, 10, and 20 mm in diameter) embedded at different depth locations, as shown in Fig. 2. Each inclusion was positioned at 6-o'clock when viewed from the top or peak looking downward, and had a stiffness contrast of approximately 4:1 compared to the healthy tissues, as described by Kashif *et al.*²³

Imaging of breast phantoms was performed with the DIET clinical prototype.¹² To ensure robust tracking of the surface motion, fiducial markers were applied on the surface randomly, as seen in Fig. 2(a) and sinusoidally actuated with 0.5 mm amplitude over a range of 10–50 Hz with a step size of 1 Hz. Prior to imaging, the cameras are calibrated by comparing the images with a known pattern and sized calibration cube. This calibration process is necessary to identify the position of the cameras in 3D space for providing 3D motion

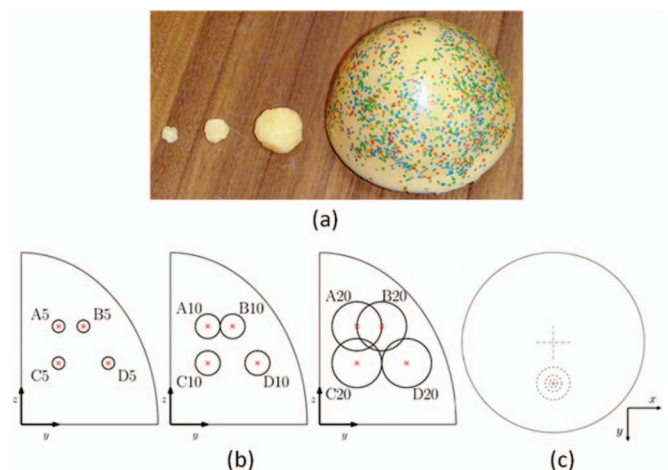


FIG. 2. (a) Silicone breast phantom with fiducial markers on the surface, required for robust tracking (Ref. 15) and examples of the stiffer inclusions placed inside, (b) the four locations (center points shown by x) for the four tumor sizes (circles), (c) tumor location B at the 6 o'clock position in the breast when viewed from the front.

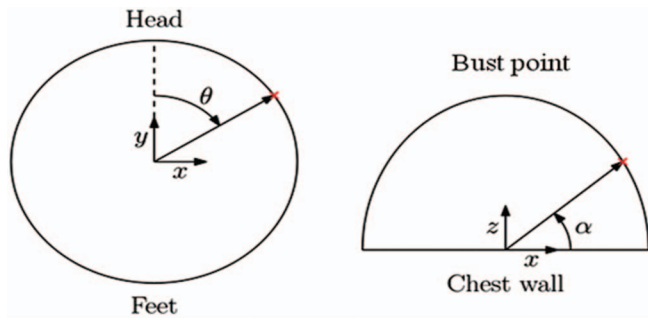


FIG. 3. Definition of angles α and θ . Angle α is the elevation angle with 0° being at the chest wall and 90° at the bust-point (nipple), and θ is the rotational angle, starting with 0° at 12-o'clock position when viewed from the front, rotating clockwise.

data of every tracked marker.^{10,15} The tracked motion paths resemble an elliptical path in 3D space. A sine curve as in Eq. (1), is fit to these paths to determine the amplitude, A , and phase, φ , in 3D for every marker displacement $\bar{x}(t)$ ¹³

$$\bar{x}(t) = \bar{A} \sin(\omega t + \bar{\varphi}), \quad (1)$$

where $\bar{A} = [A_x, A_y, A_z]^T$ and $\bar{\varphi} = [\varphi_x, \varphi_y, \varphi_z]^T$. Similar to the analysis in Ref. 19, this analysis uses the displacement data in the direction of actuation, characterized by A_z and φ_z .

A spherical coordinate system was defined, as shown in Fig. 3. Angle α is the elevation angle, with 0° being at the chest wall and 90° at the bust-point. Angle θ is the rotational angle, starting with 0° at the 12-o'clock position when viewed from front. The whole breast phantom was divided into 126 segments in terms of α and θ , to account for uneven marker coverage and the mean values of all tracked markers were used to analyze each segment. Bode diagrams of the frequency response with respect to the sinusoidal actuation were obtained for each segment.

2.B. Modal parameter estimation

The equation of motion for a linear system with n degrees of freedom (DOF) can be defined

$$M\ddot{q}(t) + B\dot{q}(t) + Kq(t) = f(t), \quad (2)$$

where $q(t)$ is the displacement vector of length n , $f(t)$ the load vector of length n , and M , B , and K are $n \times n$ mass, damping, and stiffness matrices. Assuming, proportional damping, Eq. (2) can be decoupled using the matrix of eigenmodes Uo and the coordinate transformation $q(t) = Uo p(t)$ with mass normalized eigenmodes ($Uo^T M Uo = I$) to yield

$$I\ddot{p}(t) + 2\Xi\Omega_0\dot{p}(t) + \Omega_0^2 p(t) = Uo^T f(t), \quad (3)$$

where Ξ is a diagonal matrix with dimensionless modal damping factors ζ_k , and Ω_0 is a diagonal matrix with undamped natural frequency ω_{Ok} noting that it is Ω_0^2 . Given harmonic excitation and response, the frequency response is obtained

$$\begin{aligned} \hat{q}(\omega) &= Uo(-\omega^2 I + 2j\omega\Xi\Omega_0 + \Omega_0^2)^{-1} Uo^T \hat{f}(\omega) \\ &= \sum_{k=1}^n \frac{u_{Ok} u_{Ok}^T}{-\omega^2 + 2j\zeta_k \omega_{Ok} \omega + \omega_{Ok}^2} \hat{f}(\omega). \end{aligned} \quad (4)$$

For a single-input-single-output (SISO), MDOF system, considering only the first two natural frequencies yields

$$\begin{aligned} \hat{q}(\omega) &= \frac{A1}{-\omega^2 + 2\zeta_1 \omega_{O1} \omega j + \omega_{O1}^2} \hat{f}(\omega) \\ &+ \frac{A2}{-\omega^2 + 2\zeta_2 \omega_{O2} \omega j + \omega_{O2}^2} \hat{f}(\omega), \end{aligned} \quad (5)$$

where $A1$ and $A2$ are constants determined by the eigenmodes of the system.

In this way, the problem of fitting the data in a frequency response or Bode diagram can be reduced to fitting each natural frequency independently. Quality of model fits is assessed by RMS error in estimated amplitudes $\hat{A}_i$ ($i = 1, 2$)

$$e = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{A}_i - A_i)^2}. \quad (6)$$

The frequency domain data were split into two parts and fit independently for each mode. First, the end of the first peak was determined, under the assumption that the end of this peak was characterized by a local minimum in the magnitude and a phase of 180° . The least squares fitting procedure described in Ref. 19 was used to determine the first natural frequency, ω_{O1} , and the corresponding damping factor, ζ_1 , using data up to the end of the first peak. The second natural frequency was identified using the data for frequencies larger than the first natural frequency, ω_{O1} , and the entire model structure, as defined in Eq. (5), was used to assure a good match in amplitude and phase.

2.C. Statistical analysis

The analysis was performed for 18 segments in the θ -direction, and 7 segments in α -direction, moving from 15° to 45° to avoid areas just around the actuator and around the chest wall with dominant boundary conditions. Figure 4 shows an example of the segments used overlaid upon a triangulation of the tracked fiducial markers. The two resonant frequencies identified from data within each segment were compared to those found for all other segments defined on the surface of the breast to detect significant changes. It is

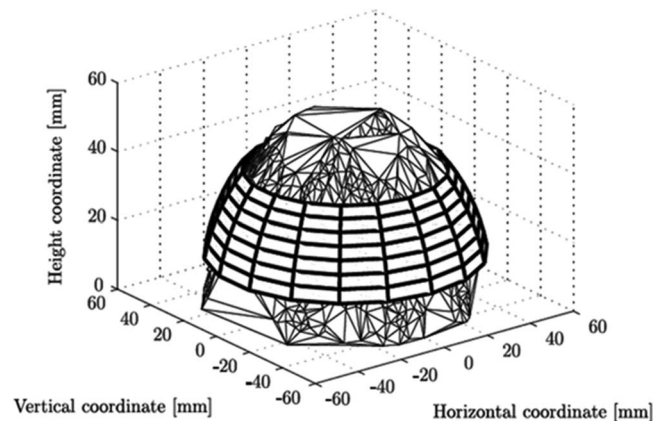


FIG. 4. An example of the used segments, overlaid upon a triangulation of the tracked fiducial markers.

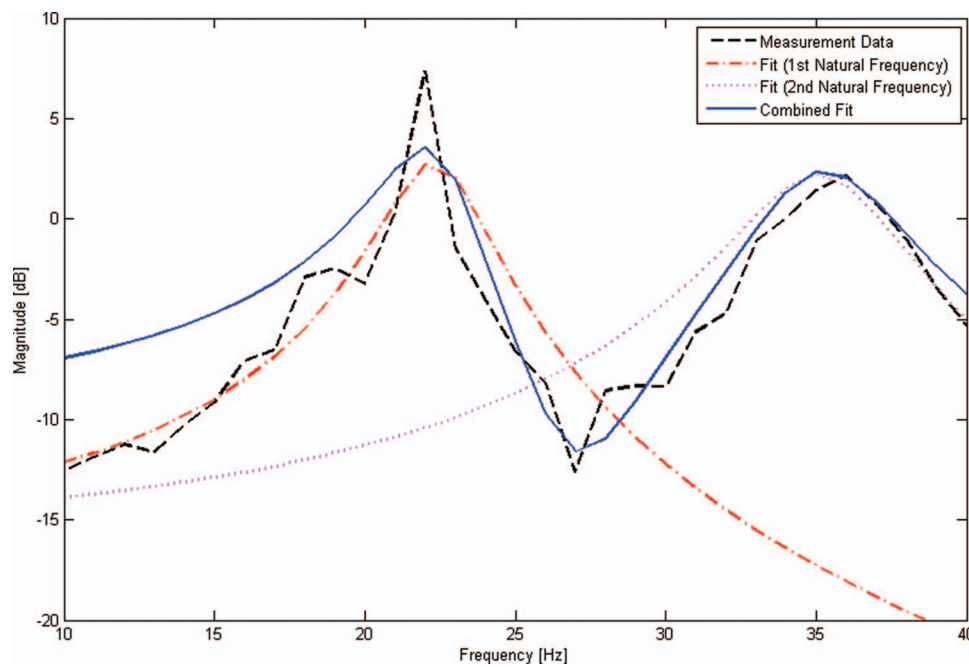


FIG. 5. A typical example of the fitting results. Measurement data are shown in dashed, fit of the first natural frequency in dashed-dotted, fit of the second natural frequency in dotted, and the combined fit is shown in solid line.

hypothesized on prior results¹⁹ that regions that have statistically significant differences in natural frequency will be located near or on inclusions and that a healthy phantom with no inclusion will show no significant differences.

To assess statistical significance of the differences observed, a two-sample *t*-test was used to reject the hypothesis that two defined regions (of 126) were similar in modal parameters. A significance threshold of $\alpha = 0.01$ was chosen, indicating a 99% confidence interval. The specific hypothesis tested was that a given area of interest has the same or different (lower) second resonance compared to all the other segments on the surface. A segment with $p < 0.01$, indicates a significantly lower second natural frequency in that section.

To analyze and detect any significant changes of this metric within the breast, statistical hypothesis testing was applied. The segments of the breast were divided into two groups. The first group included all segments within a 60° range in θ -direction. The second group consisted of all the other segments. This was repeated for all 18 θ -segments.

A two-sample *t*-test was used to compare both groups, where the null hypothesis was that both groups were from the same normal distributions with equal means and unequal variances. The alternative hypothesis was that the first group was from a different normal distribution with a lower mean value. An α -level of 0.01 was chosen, indicating a 99% confidence rate.

To furthermore assess statistical significance, the probability of obtaining the data in the analyzed segment, was calculated and called the *p*-value of the test. A segment with a low *p*-value, specifically a *p*-value below the α -level, indicated a lower second natural frequency in that section, which is an effect that has been observed to occur around the area of a stiffer inclusion.

3. RESULTS

3.A. First natural frequency, ω_{o1}

A typical fitting result for one section is shown in Fig. 5. Separate fits for the first and second natural frequency are shown, the sum of these fits produces a close match to the original measurement data. Table I shows the normalized sum squared errors, *e*, for each phantom.

In Fig. 6(a), a segmented breast viewed from the front, is overlaid with a colormap of the first natural frequency, ω_{o1} , for phantom B20 in Fig. 2(b). At the inclusion location (6-o'clock), several regions with a significantly higher first natural frequency are clearly visible, with an increase of about 3 Hz. This outcome is the effect reported in Ref. 19 and occurs in all phantoms with tumors that are 20 mm in diameter.¹⁹

TABLE I. Normalized sum squared errors, *e*, for each phantom.

Phantoms	Model (mm)
A5	7.4817
A10	7.7366
A20	7.2191
B5	5.0808
B10	8.0863
B20	9.2182
C5	8.3577
C10	6.4990
C20	6.7116
D5	5.2794
D10	7.5273
D20	11.0663
Healthy	5.0551

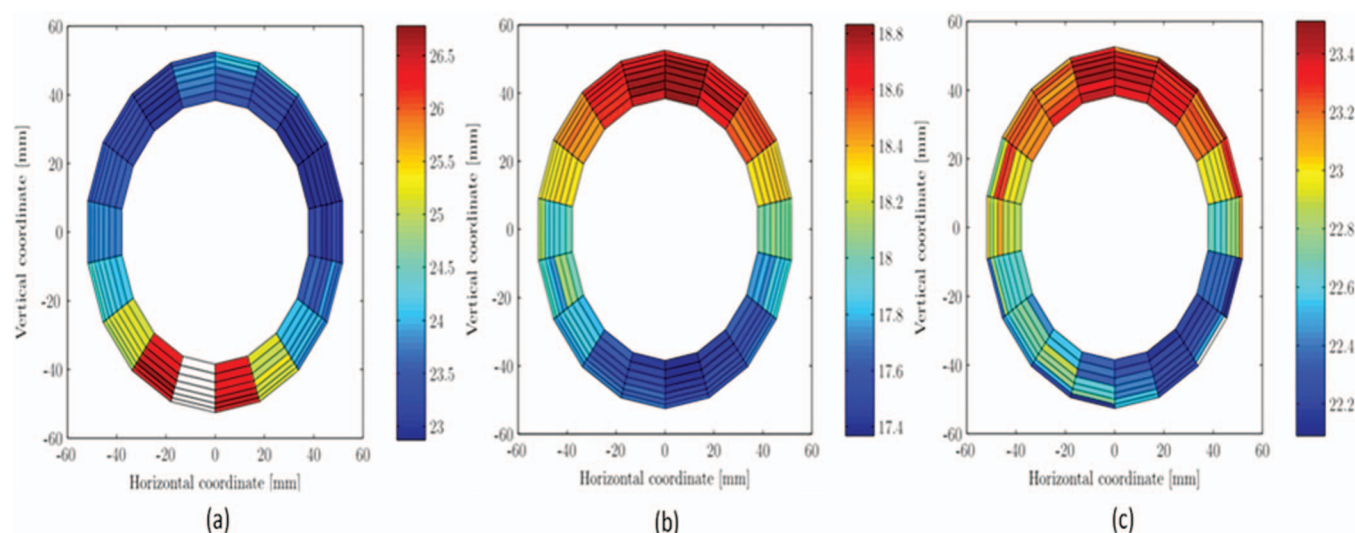


FIG. 6. Segmented breast when viewed from the front with a colormap of the first natural frequency, ω_{01} , in Hz (a) for a B20 silicone phantom, (b) for a healthy silicone breast phantom, and (c) for a B10 silicone phantom as indicated in Fig. 2(b). Blank sections indicate bad model fits due to noise and gaps in the data.

In Fig. 6(b), the homogeneous, healthy phantom has no such clear, distinctive behavior, and only a slight gradual variation of about 1 Hz is visible. In phantoms with smaller inclusions, for instance, B10 in Fig. 6(c), no distinctive first mode behavior can be found. The first natural frequency is analyzed for all phantoms and the results can be found in Figs. A1–A13, see the supplementary material in Ref. 27 for first natural frequency of all 13 phantoms.

3.B. Second natural frequency, ω_{02}

Figure 7 shows the second natural frequency, ω_{02} , for phantom B10. At the 6-o'clock, there is a lower second natural frequency clearly visible. Similar effects are visible in all phantoms with 10 and 20 mm tumor like inclusions in

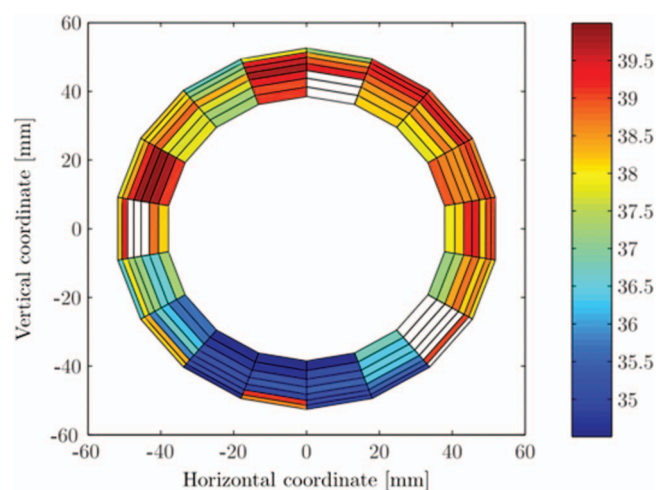


FIG. 7. Segmented breast when viewed from the front with a colormap of the second natural frequency, ω_{02} , in Hz for a B10 phantom as indicated in Fig. 2(b). Blank sections indicate bad model fits due to noise and gaps in the data.

Figs. B6–B13, see the supplementary material in Ref. 27 for analysis on the second natural frequency for all 13 phantoms. These effects are not visible in the phantoms without inclusion and not always in the phantoms with 5 mm inclusions.

Figure 8 shows the probability density functions for the second natural frequency for two different regions of phantom B10 compared to the remainder of the phantom. The black lines represent the distributions for the range $300^\circ < \theta < 360^\circ$ (top of the breast) and the red lines represent the distribution for the range $160^\circ < \theta < 220^\circ$ around the location of the inclusion. The solid lines represent the data in the given ranges, the dashed lines correspond to all the data outside this range. Clearly, the data within $60^\circ < \theta < 220^\circ$ (solid red) have a different mean frequency than the data outside this region (dashed red).

Figure 9(a) shows the statistical t -test results for ω_{02} for phantom B10. The red area (value 1) is the area where the null hypothesis is rejected, indicating a lower second natural frequency with a confidence level of 99%. The average p -value in the red area equals 6.486×10^{-7} , while the average p -value outside this area is 0.8521. The results for all 13 phantoms and the p -values in the area where the null hypothesis was rejected were compared to the p -values in the area where the null hypothesis was accepted by taking the average p -value within the stated areas. These results can be found in Table II.

All discussed results show similar behavior in all phantoms with 10 and 20 mm tumor like inclusions. Furthermore, the “healthy” phantom [t -test results shown in Fig. 9(b)] does not show this response. Results for the 5 mm inclusions remain inconclusive under this method. Figure 9(c) shows the results for phantom D5, yielding a positive result but in a false location. Figures B1–B13 have all the results of the t -test for all analyzed phantoms, see the supplementary material in Ref. 27 for analysis on the second natural frequency.

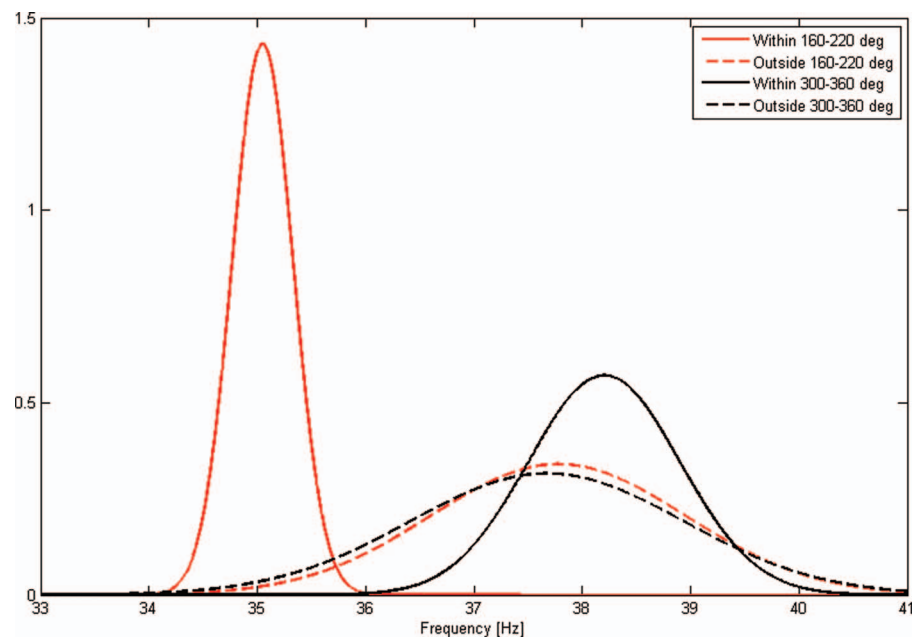


FIG. 8. Probability density function for the normal distributions in the two different segments of the breast of a B10 phantom. The light solid line represents the data in the segment $160^\circ < \theta < 220^\circ$ and the light dashed line corresponds to data outside this segment. Similarly, the dark solid line show data in the segments within $300^\circ < \theta < 360^\circ$, while the dark dashed line considers the data outside this range.

4. DISCUSSION

Phantoms are used to provide an initial proof of concept on a highly repeatable “subject.” These phantoms are anthropomorphic and validated in prior work.²³ It is important to note that before human trials can begin to fully validate this approach, initial proof of concept is an absolute necessity. In particular, such testing is limited by the ethics issues addressing burden and benefit with an unproven technology approach being tested on highly vulnerable, recently diagnosed breast cancer subjects. Hence, the use of phantoms to publish a proof of concept and method is a necessary step.

The proposed method produced good model fits to the data. By splitting the problem in two parts, the method proved very robust and reliable. Overall, the good fit in Table I indicates that a linear response of two modes can capture the fundamental dynamics, particularly natural frequency location (Fig. 5). As described in Ref. 19, the first natural frequency is a reliable metric to detect inclusions of 20 mm in diameter. That result is repeated here. In this study, the second natural frequency shows a distinctive decrease around the inclusion location, accurately detecting 20 and 10 mm inclusions, and shows the desired no change result for the homogeneous, healthy phantom. Detection of a 10 mm inclusion is clinically significant

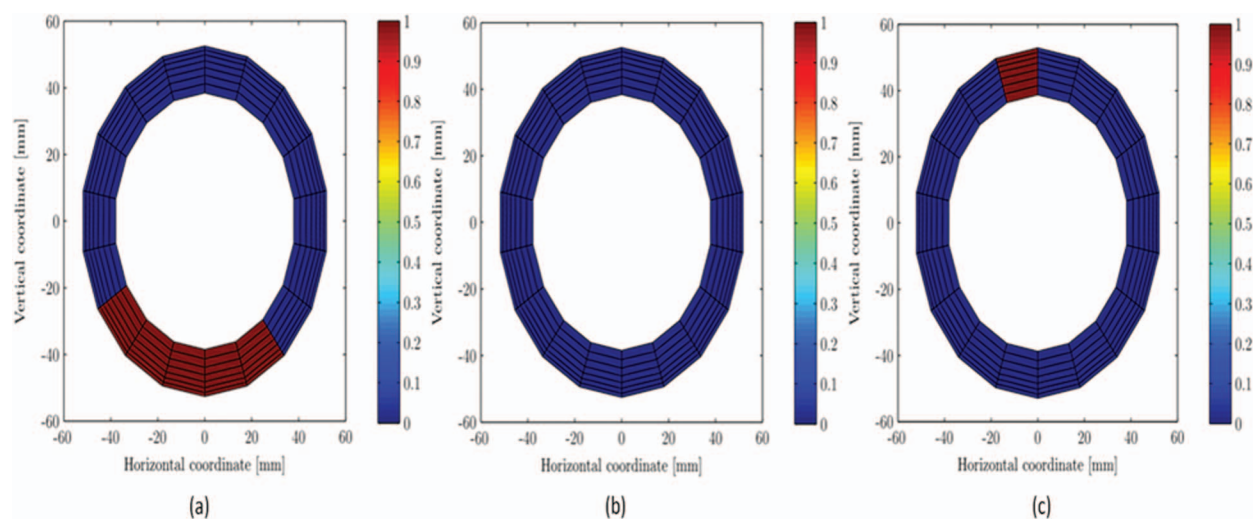


FIG. 9. Results of the t -test, (a) for the B10 phantom, (b) for the healthy phantom, and (c) for the D5 phantom, overlaid on the segmented breast. The test outcome is presented where the lighter area indicates rejection of the stated null hypothesis, indicating a lower second natural frequency and thus a stiffer inclusion, and the darker area represents acceptance of the null hypothesis.

TABLE II. Results of the t -test in terms of p -values for all 13 phantoms. A comparison is made between the segments where the null hypothesis was rejected [red area in Fig. 9(a)], possibly an indication for an inclusion, and the segments where the null hypothesis was accepted [blue area in Fig. 9(a)].

Phantoms	Average p -value in rejected area	Average p -value in accepted area
Healthy	...	0.50653
A5	4.5213×10^{-4}	0.67531
B5	1.6714×10^{-3}	0.63221
C5	1.5228×10^{-3}	0.57836
D5	7.4005×10^{-3}	0.50929
A10	1.5059×10^{-3}	0.89132
B10	6.7370×10^{-7}	0.80685
C10	2.5711×10^{-3}	0.68937
D10	3.7114×10^{-3}	0.57493
A20	1.5344×10^{-3}	0.72851
B20	1.5399×10^{-4}	0.87132
C20	4.3793×10^{-4}	0.66437
D20	2.2566×10^{-8}	0.63102

in a screening application.²⁰ Tumor like inclusions of 5 mm in diameter show no clear distinctive behavior.

Reliability of motion data is limited by boundary conditions near the actuator and chest wall, as the definition of motion in those segments is affected by the influence of the actuator and chest muscle. The randomly applied fiducial markers also cause nonavailability of motion data in certain areas near inclusion location, in the form of blank or empty segments. The likely remedy for these empty segments can be achieved by applying fiducial markers, more evenly and uniformly. This offset can also be met by improving the performance of motion reconstruction algorithms required to interpret surface motion in 3D. Moreover, imaging the breast up to higher frequencies and incorporating the third natural frequency might provide conclusive results for even smaller inclusions.

A further limitation of using phantoms is that they are never 100% representative. The human breast is a more heterogeneous structure of fat, fibroglandular tissue, and other more dense tissues. Such variation may thus affect the clarity of spatial accuracy of the modal analysis method presented. Thus, these very initial proof of concept results remain to be validated in human subjects based on these results showing that it has significant potential to work and be optimized in that clinical test environment.

However, it is clear from Fig. B2 (see the supplementary material in Ref. 27 for analysis on the second natural frequency) that the A5 case was accurately detected in this method. Hence, the lack of resolution may require more modes or may be due to the very small inclusion size not affecting modal properties over regions of the sized used here.

The distinctive behavior observed is verified by statistical analysis, with a less than 1% chance that this behavior is caused by chance. The fact that the results are consistent for all 10 and 20 mm inclusions further indicates that the results are not coincidental. Hence, the statistical approach adds robustness to the simple comparison seen in Fig. 7.

For some of the phantoms, the identified inclusion area shows a small deviation from the exact inclusion location. This deviation is a known effect and is caused by asymmetrical imaging of the particular phantom. FEM simulations of exactly the same situation, with mesh based on measured surface points of the phantom, show a similar deviation in the disturbance of the motion data. Deviations up to 45° are not a typical.^{17,18,24–26}

Furthermore, in the outcome of the t -test, a difference in size of the region where the null hypothesis was rejected, is notable. Presumably, this effect can provide a further lead for the location of the tumor inside the breast in terms of distance from the surface. Phantoms D10 and D20, where the tumor like inclusion was closest to the analyzed part of the surface (Fig. 2), show smaller regions of rejection, which might be caused by a stronger localized effect due to an inclusion very close to the surface. Inclusions deeper inside the breast (A10 and A20) show larger regions of rejection, possibly due to the fact that deeper tumors affect larger areas on the surface of the breast. Further research is necessary to assess the validity of this phenomenon, both *ex vivo* and *in vivo*. This analysis only uses the surface motion in the direction of actuation. However, the surface of the breast moves in 3D-space. Therefore, multidimensional models might prove more useful to further enhance the tumor detection algorithm for the DIET system.

It may be noted here that the malignant lesions typically are spiculated masses while fibroadenomas are mostly of spherical shapes. The inclusions embedded in the anthropomorphic breast phantoms used in this study were largely spherical which may not be a true representation of the exact shape of the cancerous lesions, but they do capture the similar impact on regional stiffness, as required for the modal analysis of the breast stiffness. It would be interesting to see the results of a phantom with two inclusions, each representing the shape and elastic properties of benign and malignant tumors, in a future study. This technique of breast cancer tumor detection however needs further validation through clinical testing, once the proof of the concept results is approved.

Overall, the concept needs further *in vivo* research to validate the results in human breasts. It is expected that there is a limit on the tumor size that can be detected with the DIET system. However, manual analysis on the measurement data shows small disturbances in the measurement data for even the 5 mm inclusions that are not consistently detected by this algorithm. Further research should focus on finding automated analysis algorithms to consistently and robustly detect these smaller inclusions.

5. CONCLUSIONS

The modal analysis approach presented is capable of detecting 10 mm diameter inclusions or larger in silicone breast phantoms in an objective and repeatable fashion, and is computationally inexpensive. However, detection of 5 mm inclusion remains inconclusive. This tumor detection approach also seems to detect the approximate location of a stiffer inclusion, which is an area of difficulty in mammography due to the compressed breast shape. This method provides a

software based, readily automated diagnosis without human intervention, enabling low operator skill requirements. Overall, this approach shows potential to provide a screening modality, complementary to mammography and other currently used screening techniques, and the results seem to justify further proof of the concept testing on more complex breast phantoms and perhaps animal data before moving to human subjects.

- ^{a)} Author to whom correspondence should be addressed. Electronic mail: amersohail786@gmail.com
- ¹ S. S. Coughlin and D. U. Ekwueme, "Breast cancer as a global health concern," *Cancer Epidemiol.* **33**(5), 315–318 (2009).
- ² *Cancer Patient Survival Covering the Period 1994 to 2003* (New Zealand Health Information Services (NZHIS), Ministry of Health, Wellington, 2006) (available URL: <http://www.nzhis.govt.nz>).
- ³ D. B. Kopans, *Breast Imaging*, 3rd ed. (Lippincott, Philadelphia, 2007).
- ⁴ M. Kriege, C. T. Brekelmans, C. Boetes, P. E. Besnard, H. M. Zonderland, I. M. Obdeijn, R. A. Manoliu, T. Kok, H. Peterse, M. M. Tilanus-Linthorst, S. H. Muller, S. Meijer, J. C. Oosterwijk, L. V. Beex, R. A. Tollenaar, H. J. de Koning, E. J. Rutgers, and J. G. Klijn, "Efficacy of MRI and mammography for breast-cancer screening in women with a familial or genetic predisposition," *New Engl. J. Med.* **351**(5), 427–437 (2004).
- ⁵ R. Fazel, H. M. Krumholz, Y. Wang, J. S. Ross, J. Chen, H. H. Ting, N. D. Shah, K. Nasir, A. J. Einstein, and B. K. Nallamothu, "Exposure to low-dose ionizing radiation from medical imaging procedures," *New Engl. J. Med.* **361**(9), 849–857 (2009).
- ⁶ A. Asghari and M. K. Nicholas, "Pain during mammography: The role of coping strategies," *Pain* **108**(1–2), 170–179 (2004).
- ⁷ P. C. Gotzsche, O. J. Hartling, M. Nielsen, J. Brodersen, and K. J. Jorgensen, "Breast screening: The facts—or maybe not," *BMJ* **338**, b86–446–448 (2009).
- ⁸ A. Sarvazy, V. Egorov, J. S. Son, and C. S. Kaufman, "Cost-effective screening for breast cancer worldwide: Current state and future directions," *Breast Cancer* **1**, 91–99 (2008).
- ⁹ L. Yang, D. M. Parkin, J. Ferlay, L. Li, and Y. Chen, "Estimates of cancer incidence in China for 2000 and projections for 2005," *Cancer Epidemiol. Biomarkers Prev.* **14**(1), 243–250 (2005).
- ¹⁰ R. G. Brown, "Three-dimensional motion capture for the DIET breast cancer imaging system," Ph.D. thesis, Department of Mechanical Engineering, University of Canterbury, 2008.
- ¹¹ J. G. Chase, C. E. Hann, and L. A. Ray, "Global motion invariant signature for fast and accurate motion tracking in a digital image based elastotomography system," U.S. patent 8249691 B2 (21 August 2012).
- ¹² T. Lotz, A. Kashif, S. Feng, P. Biret, Y. Denais, D. Lottin, L. Maillard, T. Tirschler, and J. G. Chase, "A clinical prototype of the digital image elastotomography breast cancer screening system," in *Proceedings of the 5th International Conference on Bioinformatics and Biomedical Engineering (iCBBE)*, Wuhan, China, 2011.
- ¹³ T. Lotz, P. D. Simpson, D. Stocker, C. E. Hann, and J. G. Chase, in *Proceedings of the 32nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS)*, Buenos Aires, Argentina, 2010.
- ¹⁴ A. Samani, J. Zubovits, and D. Plewes, "Elastic moduli of normal and pathological human breast tissues: An inversion-technique-based investigation of 169 samples," *Phys. Med. Biol.* **52**(6), 1565–1576 (2007).
- ¹⁵ R. G. Brown, C. E. Hann, J. G. Chase, and L. E. Ray, "Discrete colour-based euclidean-invariant signatures for feature tracking in a DIET breast cancer screening system," *Proc. SPIE* **6511**, (2007).
- ¹⁶ R. G. Brown, C. E. Hann, and J. G. Chase, "Vision-based 3D surface motion capture for the diet breast cancer screening system," *Int. J. Comput. Appl. Technol.* **39**(1), 72–78 (2010).
- ¹⁷ A. Peters, J. G. Chase, and E. E. Van Houten, "Digital image elastotomography: Mechanical property estimation of silicone phantoms," *Med. Biol. Eng. Comput.* **46**(3), 205–212 (2008).
- ¹⁸ A. Peters, J. G. Chase, and E. E. Van Houten, "Digital image elastotomography: Combinatorial and hybrid optimization algorithms for shape-based elastic property reconstruction," *IEEE Trans. Biomed. Eng.* **55**(11), 2575–2583 (2008).
- ¹⁹ T. F. Lotz, N. Muller, C. E. Hann, and J. G. Chase, "Minimal elastographic modeling of breast cancer for model based tumor detection in a digital image elastotomography (DIET) system," *Proc. SPIE* **7963**, 796322–796326 (2011).
- ²⁰ J. S. Michaelson, M. Silverstein, J. Wyatt, G. Weber, R. Moore, E. Halpern, D. B. Kopans, and K. Hughes, "Predicting the survival of patients with breast carcinoma using tumor size," *Cancer* **95**(4), 713–723 (2002).
- ²¹ P. Götzsche, K. Jørgensen, P.-H. Zahl, and J. Mæhlen, "Why mammography screening has not lived up to expectations from the randomised trials," *Cancer Causes Control* **23**(1), 15–21 (2012).
- ²² L. Tabar, G. Fagerberg, H.-H. Chen, S. W. Duffy, C. R. Smart, A. Gad, and R. A. Smith, "Efficacy of breast cancer screening by age: New results from the Swedish two-county trial," *Cancer* **75**(10), 2507–2517 (1995).
- ²³ A. S. Kashif, T. F. Lotz, M. D. McGarry, A. J. Pattison, and J. G. Chase, "Silicone breast phantoms for elastographic imaging evaluation," *Med. Phys.* **40**(6), 063503 (11pp.) (2013).
- ²⁴ A. Peters, A. Milsant, J. Rouze, L. E. Ray, J. G. Chase, and E. E. Van Houten, "Digital-image based elastotomography: Proof of concept studies for surface based mechanical property reconstruction," *JSME Int. J.* **47**(4), 1117–1123 (2004).
- ²⁵ A. Peters, S. Wortmann, R. B. Elliot, M. Staiger, J. G. Chase, and E. E. Van Houten, "Digital image-based elastotomography: First experiments in surface based mechanical property estimation of gelatine phantoms," *JSME Int. J.* **48**(4), 562–569 (2005).
- ²⁶ A. Peters, H.-U. Berger, J. G. Chase, and E. E. W. Van Houten, "Digital image-based elastotomography: Nonlinear mechanical property reconstruction of homogeneous gelatine phantoms," *Int. J. Inf. Syst. Sci.* **2**(4), 512–521 (2006).
- ²⁷ See supplementary material at <http://dx.doi.org/10.1181/1.4826168> for Figs. A1–A13 and B1–B13.